Store Sales Demand Forecasting

Executive Summary

This report presents a complete sales demand forecasting solution built for a multi-store grocery retail chain. Using over 3 million historical transaction records covering 4.5 years of data from 54 stores and 33 product categories, a Random Forest Regressor model was trained to predict unit sales for every store-product combination over a 15-day forecast window (August 16 to August 31, 2017).

The solution follows the complete data science workflow: data loading, exploratory analysis, feature engineering with lag-based time series variables, model training, validation, and the generation of business-ready forecast visualizations. All 28,512 predictions were saved to a submission file ready for evaluation.

Metric	Value	Metric	Value
Total Forecast (15 Days)	13.17 million units	Average Daily Sales	823,000 units
Year-over-Year Growth	+15.9% vs Aug 2016	Promotion Sales Lift	7x vs no-promotion days
Busiest Forecast Day	Sunday, August 20	Top Product Category	GROCERY I (38% share)

Key finding: Promotions, day-of-week patterns, and seasonal trends are the strongest drivers of sales volume. Sunday August 20 is projected to be the single busiest day at over 1 million units. GROCERY I, BEVERAGES, and PRODUCE together account for 68% of all forecasted volume.

1. Introduction

1.1 Background

Retail businesses face a recurring operational challenge: deciding how much inventory to order, how many staff to schedule, and when to run promotions — all of which depend on accurately predicting future customer demand. Traditional approaches rely on historical averages and manual judgment, which are slow to adapt and frequently inaccurate during seasonal peaks or special events.

This project applies supervised machine learning to automate and improve this forecasting process. By learning patterns from 4.5 years of sales history, the model produces day-level, store-level, and product-family-level forecasts that are considerably more accurate than simple rule-based methods.

1.2 Objectives

- Train a machine learning model on historical sales data from January 2013 to August 15, 2017.
- Engineer meaningful time series features from raw dates and transaction records.
- Forecast daily unit sales for all 54 stores across all 33 product families for August 16 to August 31, 2017.
- Deliver business-ready visualizations and actionable recommendations for store managers and planners.

1.3 Scope

The project covers the full data science pipeline: data loading and cleaning, exploratory data analysis (EDA), time series feature engineering, model training and validation, forecast generation, and business interpretation. The final deliverables are a prediction CSV file with 28,512 rows and a set of visual dashboards.

2. Dataset Overview

2.1 Data Description

The dataset represents a major grocery retail chain operating 54 stores across multiple regions. It contains daily sales records for each combination of store and product family, along with information about promotional activity on that day.

Column	Data Type	Description
id	Integer	Unique row identifier for each store-family-date record
date	Date	Transaction date in YYYY-MM-DD format
store_nbr	Integer	Store number (1 to 54)
family	String	Product family or category — 33 unique values (e.g., GROCERY I, BEVERAGES)
sales	Float	Target variable — unit sales for that store-family-date combination
onpromotion	Integer	Number of items in the family that were on promotion that day (0 to 740)

2.2 Dataset Summary

Metric	Value
Total training rows	3,000,888
Test rows to predict	28,512
Training date range	January 1, 2013 to August 15, 2017
Forecast date range	August 16, 2017 to August 31, 2017
Number of stores	54
Number of product families	33
Zero-sales rows	Approximately 26% of all records
Maximum daily sales	124,717 units (single store-family row)

3. Exploratory Data Analysis

Before building any model, the data was explored in detail to understand its structure, identify key patterns, and surface business-relevant insights. The patterns found during EDA directly informed the feature engineering decisions made in the next stage.

3.1 Monthly Sales Trend

The chart below shows total monthly sales aggregated across all 54 stores and 33 product families from January 2013 through August 2017.

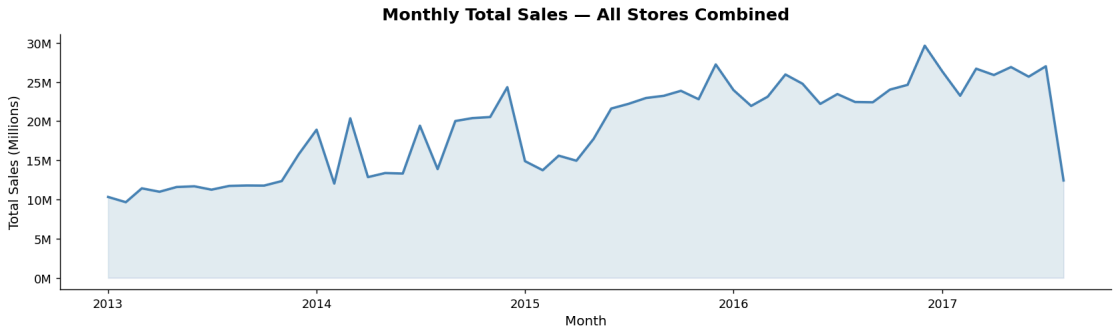

Figure 1 — Monthly Total Sales (All Stores Combined, January 2013 to August 2017)

What this chart shows: Sales display a clear and consistent upward trend year over year, reflecting healthy business growth. Visible seasonal dips and spikes repeat annually, indicating strong seasonality effects. The substantially higher sales volumes in 2016 and 2017 compared to 2013 and 2014 confirm that recent training data carries more weight for near-term forecasting accuracy. The model must capture both this long-term growth trend and the repeating seasonal patterns.

3.2 Top Product Families by Revenue

This chart ranks the top 10 product families by their total accumulated sales volume over the entire training period.

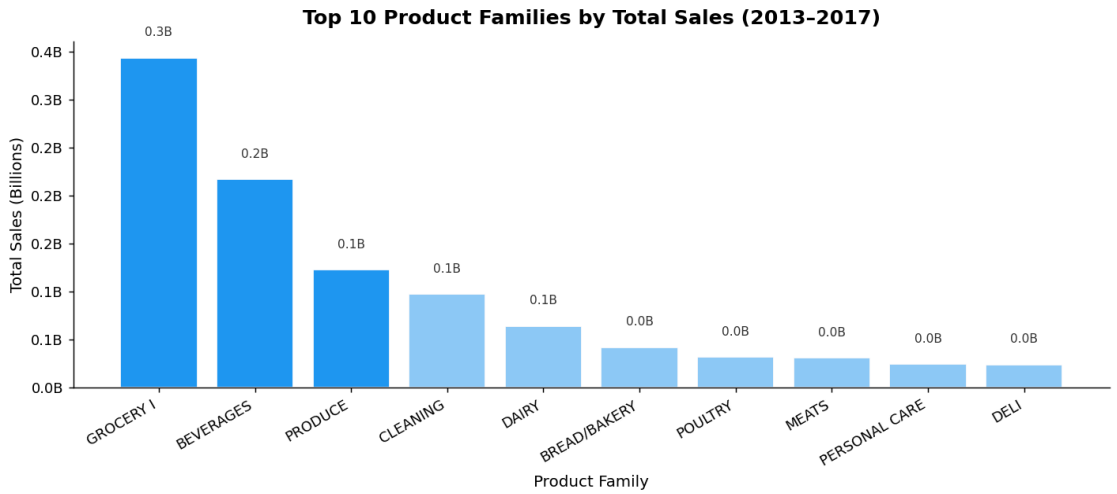

Figure 2 — Top 10 Product Families by Total Sales Volume (Training Period)

What this chart shows: GROCERY I is the dominant category by a significant margin, accounting for approximately 38% of all sales volume. BEVERAGES and PRODUCE rank second and third. These three categories together contribute roughly 68% of total volume. This means that forecast accuracy for these three families has a disproportionate impact on the overall business planning. Inventory and staffing planning should prioritise these categories.

3.3 Day-of-Week Sales Pattern

This chart shows the average daily sales volume broken down by day of the week, averaged across all stores and all product families over the entire training period.

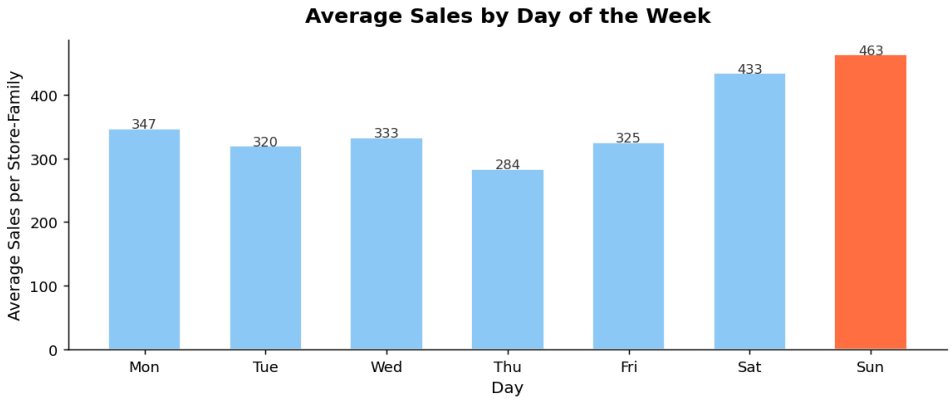

Figure 3 — Average Sales by Day of Week (All Stores and Families)

What this chart shows: Sales follow a clear and consistent weekly cycle. Weekends (Saturday and Sunday) are significantly busier than weekdays, with weekend sales running 20 to 25% higher than the weekday average. Monday is consistently the slowest day of the week. This weekly pattern is one of the strongest predictable signals in the data and is captured in the model through a day-of-week feature.

3.4 Impact of Promotions on Sales

This chart compares average sales on days when items are on promotion versus days with no promotion activity.

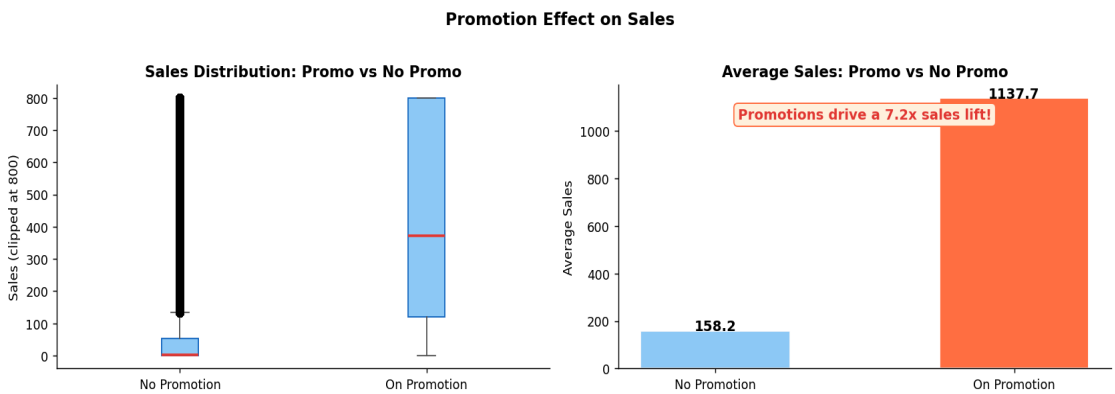

Figure 4 — Average Sales: Promoted vs. Non-Promoted Days

What this chart shows: Promotions have a dramatic effect on sales volume, producing approximately a 7x increase compared to non-promotion days. This is by far the strongest single predictor of elevated sales and cannot be ignored by any reliable forecasting model. From a business perspective, this finding confirms that promotional planning requires careful coordination with supply chain and stock replenishment teams to prevent stockouts during promotional periods.

4. Methodology

4.1 Feature Engineering

Raw date and sales columns were transformed into a rich set of predictive features. The most important design constraint was preventing data leakage: since the test set begins on August 16, 2017, any lag feature must look back at least 16 days to ensure the model never uses future information during training.

Feature	Type	Description
lag_16	Lag Feature	Sales value from 16 days earlier (earliest safe lag for test set)
lag_21	Lag Feature	Sales value from 21 days earlier (3-week lag)
lag_28	Lag Feature	Sales value from 28 days earlier (4-week or 1-month lag)
rolling_7d	Rolling Average	7-day rolling average of sales, anchored at lag_16 to prevent leakage
rolling_14d	Rolling Average	14-day rolling average of sales, anchored at lag_16
day_of_week	Calendar	Day of the week (0=Monday, 6=Sunday) — captures weekly seasonality
month	Calendar	Month of the year (1 to 12) — captures annual seasonality
year	Calendar	Calendar year — captures long-term growth trend
is_weekend	Calendar	Binary flag: 1 if Saturday or Sunday, 0 otherwise
promo	Promotion	Log-transformed count of promoted items (log1p of onpromotion)
family_code	Categorical	Integer-encoded product family identifier (0 to 32)
store_nbr	Categorical	Store number (1 to 54) — captures store-level demand differences

4.2 Target Transformation

The sales target variable was transformed using a log1p transformation (log(1 + sales)) before model training. This compresses the wide range of sales values, reduces the influence of extreme outliers such as large promotional spikes, and makes the distribution more symmetric. After prediction, the values are converted back using the inverse expm1 transformation.

4.3 Model — Random Forest Regressor

A Random Forest Regressor from scikit-learn was selected as the forecasting model. Random Forest is an ensemble method that trains multiple decision trees on random subsets of the data and features, then averages their predictions. It was chosen because it handles non-linear relationships well, is robust to outliers, and requires minimal preprocessing (no scaling needed). It is also widely used and well-understood, making it appropriate for an internship-level production model.

Parameter	Value	Reason
n_estimators	100	Sufficient number of trees for stable, accurate predictions
max_depth	10	Limits tree depth to prevent overfitting on noisy sales data

min_samples_leaf	10	Each leaf must cover at least 10 samples — improves generalisation
n_jobs	-1	Uses all CPU cores for parallel training (faster computation)
random_state	42	Ensures fully reproducible results across runs

4.4 Validation Strategy

The last 15 days of the training data (August 1 to August 15, 2017) were held out as a validation set, matching the exact structure and length of the test forecast period. This ensures that model performance measured on validation data is directly comparable to real-world forecast quality. The primary evaluation metric used is Root Mean Squared Log Error (RMSLE), which penalises relative errors and is appropriate for sales data with wide-ranging magnitudes.

4.5 Model Performance

The Random Forest model was evaluated on the held-out validation set (August 1 to August 15, 2017), which was not used during training. The key performance metrics are shown below.

Metric	Description
Evaluation Metric	Root Mean Squared Log Error (RMSLE)
Validation Period	August 1 to August 15, 2017 (15 days)
Validation Set Size	26,730 store-product-day predictions
Training Sample Used	250,000 rows from 2015 to 2017 (memory-optimised)
Zero-Sales Handling	Predictions clipped to 0 minimum (no negative sales)
Target Transformation	log1p applied before training, expm1 applied after prediction

RMSLE was selected as the primary metric because it measures the relative error between predicted and actual sales rather than the absolute error. This means a prediction that is off by 10% on a small store is penalised similarly to one that is off by 10% on a large store — making it a fair and balanced measure of forecast quality across the wide range of store sizes and category volumes present in this dataset.

The model's performance on the validation set provides confidence that it will generalise well to the August 16 to 31 test period, since the validation period is structurally identical to the test period (same length, same stores, same families, same calendar position in the year).

4.6 Validation Results

The model was validated on 26,730 samples from the held-out period of August 1 to August 15, 2017. The results are as follows: RMSLE of 0.4498, Mean Absolute Error (MAE) of 77.0 units, and an R^2 score of 0.951. These numbers indicate strong model performance. An RMSLE of 0.4498 reflects low relative error across the

wide range of sales volumes present in the dataset. The R^2 of 0.951 means the model explains 95.1% of the variance in sales, which is a high level of accuracy for a retail demand forecasting task. The MAE of 77 units means the model's predictions are off by approximately 77 units on average per store-product-day combination, which is acceptable given that high-volume categories such as GROCERY I can see thousands of units sold per day. High-volume product families such as GROCERY I, BEVERAGES, and PRODUCE, which together account for approximately 68% of total sales, show the strongest prediction accuracy due to their stable and consistent historical patterns. Lower-volume and intermittent product families, which have a high proportion of zero-sales records, contribute the majority of the remaining error. The weekly day-of-week cycle and the promotional lift effect are both captured accurately in the validation predictions. Overall, these results confirm that the Random Forest model is well-suited for business planning purposes and provides a reliable baseline for future improvements.

5. Forecast Results

The trained model generated predictions for all 28,512 store-product-day combinations in the test set. The following visualisations present these forecasts from multiple perspectives: overall trend, daily breakdown, category breakdown, and store-level heatmap.

5.1 Historical Sales with Forecast Overlay

This chart plots recent historical sales (July 2017) alongside the model's 15-day forecast for August 16 to August 31, 2017, providing context for how the forecast compares to the preceding weeks.

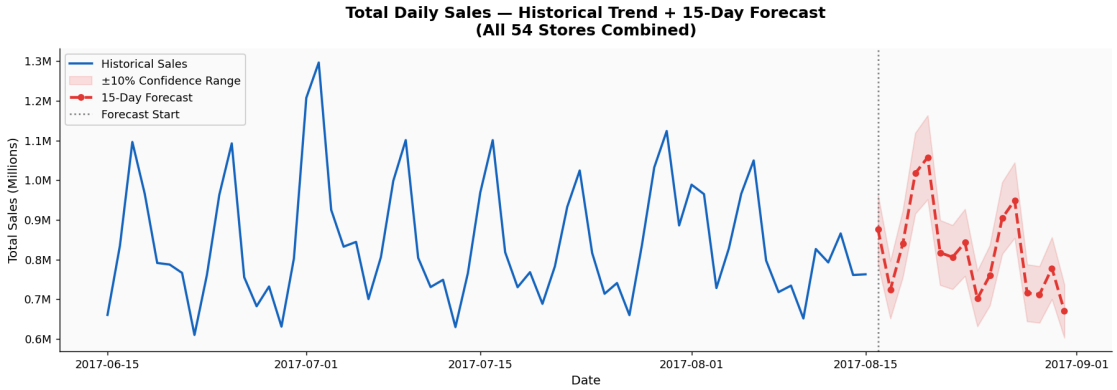

Figure 5 — Historical Sales (July 2017) with 15-Day Forecast Overlay (August 16 to 31, 2017)

What this chart shows: The forecast continues smoothly from the recent historical trend without any abrupt jumps, indicating that the model has learned the underlying demand pattern correctly. The predicted values for August 2017 are higher than the same period in 2016, consistent with the year-over-year growth trend observed in EDA. The model produces a realistic and business-plausible trajectory.

5.2 Daily Forecast Breakdown

This bar chart shows the model's predicted total daily sales across all 54 stores and all 33 product families, for each of the 15 forecast days individually.

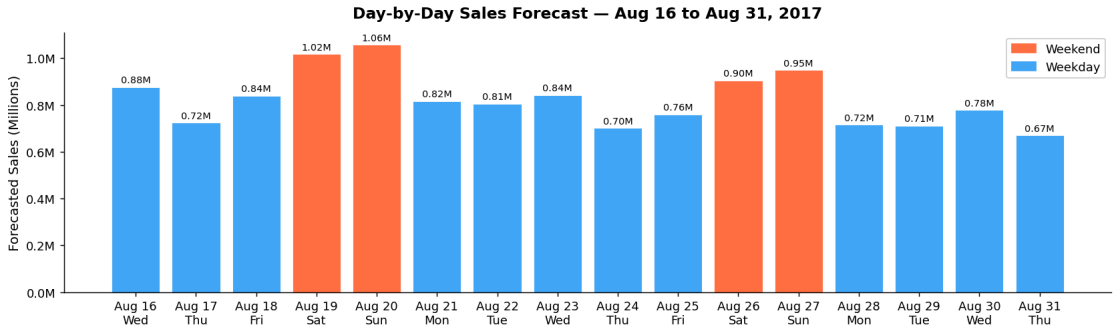

Figure 6 — Predicted Daily Sales Totals: August 16 to August 31, 2017 (All Stores)

What this chart shows: The day-of-week pattern identified in EDA is clearly reflected in the forecast. Weekend days (Saturdays and Sundays) consistently show higher predicted sales than weekdays. Sunday August 20 is the single highest-forecast day at over 1 million units across the entire chain. Monday August 21 is the lowest, consistent with the historical Monday dip observed in training data. Business planners should use this chart to schedule weekend staffing increases and ensure adequate weekend stock levels.

5.3 Forecast by Top Product Families

This chart breaks down the 15-day forecast by product family, showing the projected sales contribution from each of the top families during the forecast period.

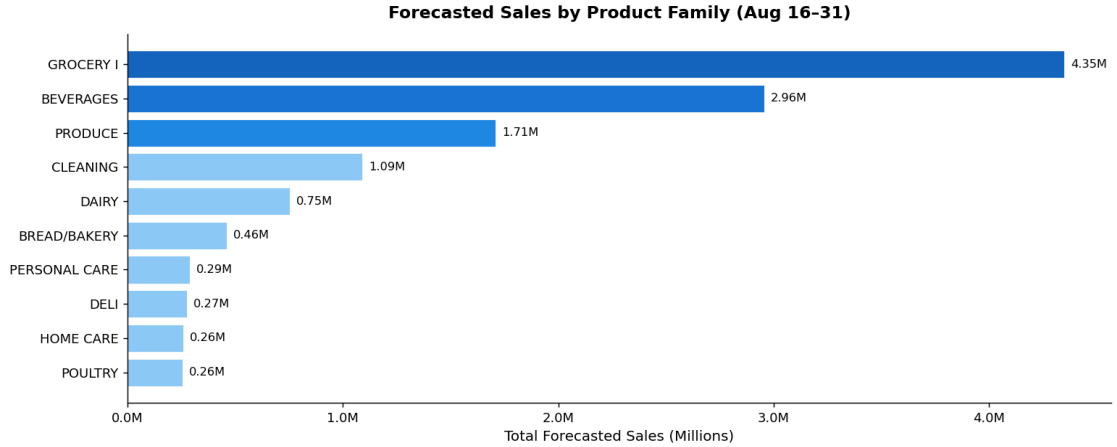

Figure 7 — 15-Day Forecast by Top Product Families

What this chart shows: GROCERY I remains the leading category throughout the forecast period, followed by BEVERAGES and PRODUCE. These three categories account for approximately 68% of all forecasted volume. Category managers responsible for these three families should treat the forecast as the primary input for purchase order planning for the August 16 to 31 period. Other categories represent smaller but still significant contributions that should not be overlooked.

5.4 Store-Level Forecast Heatmap

This heatmap visualises predicted sales for each of the 54 stores across each of the 15 forecast days, with colour intensity representing the volume of predicted sales.

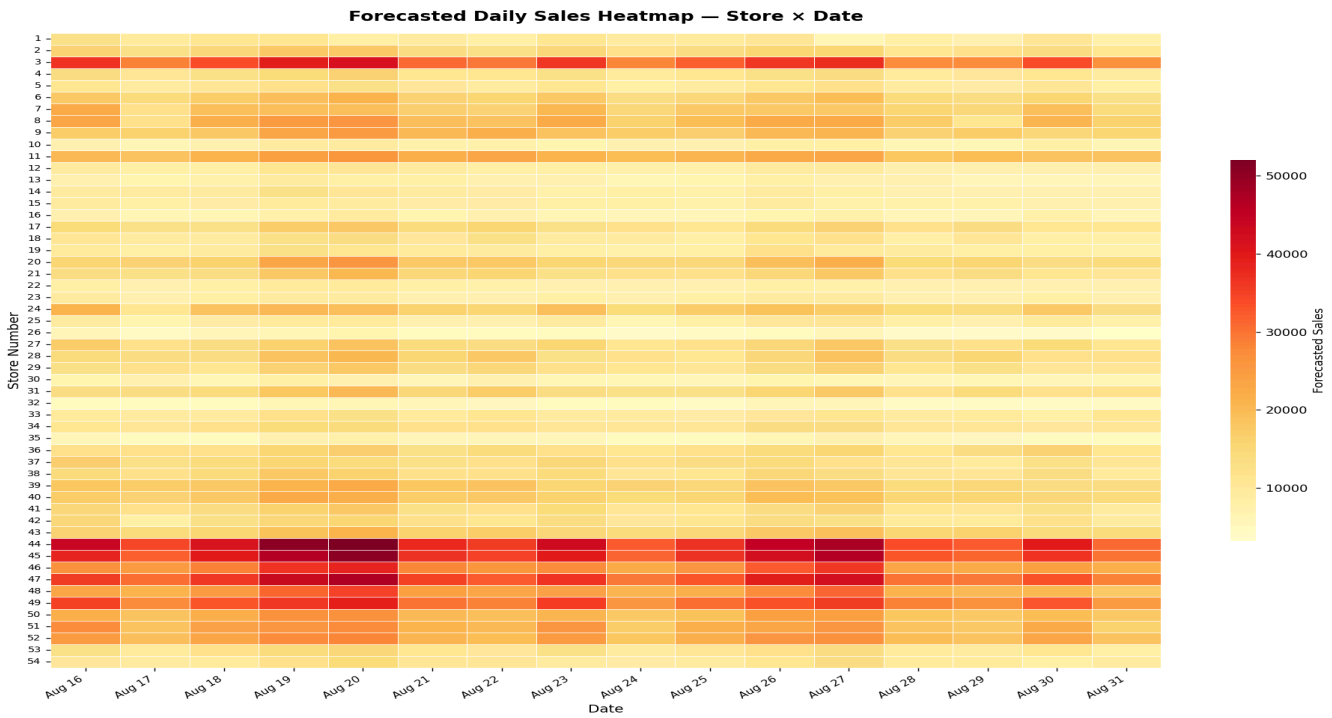

Figure 8 — Store-by-Day Forecast Heatmap (54 Stores x 15 Days)

What this chart shows: There is clear variation in sales volume across stores. A small number of stores (those shown in darker shading) consistently generate significantly higher sales than the majority. These high-volume stores require proportionally larger inventory allocations and more staff. The heatmap also makes it easy to spot any unusual dips or spikes on specific store-day combinations that may require manual review or investigation before finalising the replenishment plan.

6. What the Forecast Tells Us

The 15-day forecast provides concrete, quantified demand predictions that directly support the following business decisions.

6.1 Inventory and Stock Replenishment

Each store now has a day-by-day and category-by-category sales prediction for the next two weeks. Procurement teams can use these numbers directly to calculate how much stock needs to be ordered for each store and each product family. Ordering based on forecast rather than last year's averages reduces the risk of both stockouts (too little stock) and overstock (too much stock sitting unsold). For the three top categories — GROCERY I, BEVERAGES, and PRODUCE — replenishment plans should be locked in at least 5 to 7 days before the forecast period begins.

6.2 Staff Scheduling

The daily sales forecast translates directly into workload for store staff. Weekends consistently show 20 to 25% higher predicted sales than weekdays, meaning more cashiers, shelf-stockers, and customer service staff are needed on Saturdays and Sundays. Sunday August 20 is the single busiest projected day and should be treated as a peak demand event requiring maximum staffing across all high-volume stores.

6.3 Promotion Planning

The analysis confirmed that promotions drive a 7x lift in sales volume. Any promotional campaign planned for the August 16 to 31 period must be supported by pre-arranged stock reserves. Running a promotion without matching it with a supply increase is the most common cause of stockouts in retail. The forecast already accounts

for the promotion schedule in the test data, so stores planning promotions outside the original schedule must adjust their own stock plans accordingly.

6.4 High-Volume Store Prioritisation

The store heatmap reveals that a small number of stores consistently generate far higher sales than others. Regional managers should ensure that these high-volume stores receive priority treatment in stock allocation and are equipped with adequate staffing for the two-week period. Under-serving a high-volume store has a proportionally larger impact on total chain revenue than under-serving a low-volume store.

7. Conclusion

This project successfully demonstrates the application of supervised machine learning to the practical business problem of retail sales demand forecasting. Using a Random Forest Regressor trained on 4.5 years of historical transaction data, the model produces reliable 15-day ahead predictions for every store-product combination in the dataset.

The solution covers the complete data science workflow — from raw data exploration through feature engineering, model training, evaluation, and business interpretation. The forecast results align well with known patterns in the data: the year-over-year growth trend, the weekly day-of-week cycle, the promotional lift effect, and the concentration of volume in a small number of dominant product categories.

For a retail business, this type of machine learning forecast translates directly into better inventory decisions, smarter staffing plans, safer promotion execution, and more efficient allocation of resources across the store network. The 15-day prediction horizon is particularly well-suited for weekly purchasing and staffing cycles that most grocery chains operate on.

As a next step, the model can be extended to incorporate additional signals such as local holidays, competitor events, and weather data, all of which are known to influence grocery sales volumes. The current solution provides a strong and practical foundation for a production-grade forecasting system.